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PHY 211 — Past Questions & Materials
by Renance

Course: PHY 211
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 7
Levels: 200

This study pack was compiled and restructured by Renance for PHY 211 students at Federal University of Petroleum Resources, Effurun. It brings together 7 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 7 — PHY 211 - 2014/2015 - First Semester - ELECTRIC CIRCUIT AND ELECTRONICS

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

15th SEMESTER EXAMINATION (2014/2015 SESSION)

PHY 211'(ELECTRIC CIRCUIT AND ELECTRONICS) (3 UNITS)

Instruction: Answer any three (3) questions Time: 2Vhrs

1. (a) What is the difference between forward bias, reverse bias and zero bias in a Diode..
- (b) How does avalanche voltage compare with the forward break over voltage
- (c) Suppose two signals, one having input frequency of 3.500MHz and the other having an input frequency of 2.5MHz are fed into the input of the mixer. What are the output signal frequencies?
- (d) Suppose a single semi-conductor for PV cell can produce 100A of current in direct sunlight. How many of these would be necessary, if connected in parallel to produce 500mA?
- (e) What is the static forward current transfer ratio in a device where a base current of 3mA produces a collector current of 25mA
- (f) The acronym MOSFET stands for what?
- (g) What is the current gain in a transistor where a change in the base current of 0.1mA results in a change in the collector current of 1A
- (h) The term common drain implies that
- (i) Give three ways which an NPN transistor can be used
- (j) If the AC input voltage is 10mV and the rms AC output voltage is 70mV. What is the voltage gain? Assume the input and output impedances are identical.
2. (a) Define Semi-Conductors
- (b) With circuit diagrams and with anode and cathode symbology diagram. Explain
 - (i) Forward bias current flow
 - (ii) Reverse bias current flow
3. (a) Explain Signal Amplifier.
- (b) Draw a circuit diagram to explain Envelope Detection as an application of signal
4. (a) Explain the principle of JFET (Junction Field Effect Transistor)
- (b) With pictorial diagrams and schematic symbols
 - (i) NPN Transistors

(ii) PNP Transistors |

. (a) Explain the basic FET circuits.

* (b) With circuit diagram, explain the common source circuit configuration,

. (a) Explain Signal Oscillators

(b) With the simple diagram of the basic concept of signal oscillator. explain briefly (i) the feedback

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF PYHSICS

COURSE TITLE: ELECTRIC CIRCUITS AND ELECTRONICS

COURSE CODE: PHY 211

ACADEMIC SESSION: 2016/2017 TIME ALLOWED: 2 hours 30 minutes

INSTRUCTIONS: ANSWER ANY FOUR QUESTIONS

»* Question 1

a) Explain cut off Region and saturation point of a transistor.

b) State and explain the two mechanisms responsible for reverse junction breakdown voltage

c) A transistor connected in a common base configuration has emitter current $I_E = 5\text{mA}$, $I_C = 3.5\text{mA}$. (i) Determine the short circuit gain. (ii) Obtain an expression to show the relationship between the common emitter gain and the common base circuit gain.

Question 2

a), Explain the working principle of the Zener diode

by State the operating principle and one use each of the;

i) Light emitting Diode ~ ~"

ii) Schottky Diodes

iii, Photodiodes

c) A common emitter configuration has an output voltage $V_{CE} = 20\text{V}$, load resistance

$R_L = 20\text{k}\Omega$; $\beta_{FE} = 25$; $R_B = 10\text{k}\Omega$. (i) Calculate the input voltage V_{in} . (ii)

Output current I_{out} . (iii) Voltage gain A_v ,

/ Question 3

a) Explain the differences between the direct band gap semiconductor and the indirect band gap semiconductor.

b) State four features of an N-type semiconductor, and draw the lattice structure of silicon doped with Arsenic.

A) A Gallium Phosphide LED with a band gap of $3.62 \times 10^{-19}\text{J}$ is incident upon by a photon with energy equal to the band gap energy. Find the momentum of the photon. What colour of light will be emitted by this material?

V Question 4

a) Write a short note on the non-linearity of the diode, using the voltage/ current characteristic graph.

Aeh State the purpose of the three terminals and the channel of the Field Effect Transistor (FET)

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c) Define Power gain A_p of a transistor. if the power input to the transistor is; $P_i = (I_i R_i)$, derive a relationship between the Power gain A_p and voltage gain A_v of a common emitter transistor.

Question 5

1-2) Explain the formation of the PN junction.

2) state five applications of a diode.

c) A transistor with a common emitter configuration has a load resistance of $3.5k\Omega$, provides a voltage gain of 105, with an input resistance of $1.5k\Omega$.

Calculate the voltage and power gain of the transistor, if the load resistance is halved and the input resistance reduced to $0.5k\Omega$. Assume a constant current gain.

Question 6

1) What is a feedback system? State three importances of feedback systems.

2) Draw the block diagram of the basic elements of the feedback oscillator and state how a continuous output is obtained in an oscillator.

3) (i) Define band gap. If using Einstein's energy mass relation, explain the relationship between momentum and energy in an indirect band gap semiconductor.

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Paper 2 of 7 — PHY 211 Electric Circuit and electronics 2024/2025

Level 200 • First Semester • PastQuestion

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05/04/2024
 Anode miracle Peter

Federal University of Petroleum Resources Effurun, Nigeria.
 College of Science, Department of Physics.
 First Semester Examination

Academic Session: 2023/2024
 Course Title: Electric Circuit & Electronics
 Course Code: PHY211 (3 units)
 Time Allowed: 2½ hrs.

Date: /05/2024

Instructions: Answer Any Four (4) Questions. All Questions carry equal marks. Note that all symbols have their usual meanings.

QUESTION 1

- (a) (i) Distinguish between direct and indirect band gap of a semiconductor (ii) State one example each of both band gaps (iii) Which of this band gap is used for optical devices?
 (b) Briefly explain the following terms: (i) forbidden band gap (ii) Fermi energy (iii) Knee voltage (iv) Filter circuit (v) Avalanche breakdown
 (c) Why is Carbon unable to function as a semiconductor?
 (d) Briefly describe the conductivity of a semiconductor at 0°C and at room temperature.

(17.5 mark)

QUESTION 2

- (a) Distinguish between DONOR and ACCEPTOR impurities
 (b) Mention three examples each of P-type and N-type dopant atoms
 (c) Briefly discuss the biasing of the P-N junction diode
 (d) List and explain the two mechanisms at which the PN junction breaks down.
 (e) A crystal diode having internal resistance $r_f = 20\ \Omega$ is used for half-wave rectification. If the applied voltage is $V = 50 \sin \omega t$ and load resistance, $R_L = 800\ \Omega$, calculate (i) mean current (I_m) (ii) I_{rms} (iii) a.c. power input.

(17.5 mark)

QUESTION 3

- (a) Discuss the operation of the Full Wave Bridge Rectifier
 (b) (i) What is a filter circuit? (ii) Mention two types of filter circuits (iii) State the basic electrical principles governing the filtering action of inductors and capacitors
 (c) How does Zener diode act as a voltage stabilizer?
 (d) A full-wave rectifier uses two diodes; the internal resistance of each diode may be assumed constant at $20\ \Omega$. The transformer a.m.s secondary voltage from centre

tap to each end of secondary winding is 50 V and load resistance is 980 Ω . Find the r.m.s value of the load current.

(17.5 mark)

QUESTION 4

- (a) Define the following: (i) Transistor amplifier (ii) Current amplification factor (iii) Base-current amplification factor (iv) Collector-base current with emitter open (v) Transistor reverse biasing
(b) Derive the relation:

$$I_c = \frac{\alpha}{(1-\alpha)} I_B + \frac{I_{CBO}}{(1-\alpha)}$$

- (c) In a common base connection, the current amplification factor (α) is 0.9. if the emitter current I_E is 1mA, determine the leakage current $I_{leakage}$.

(17.5 mark)

QUESTION 5

- (a) (i) What is the purpose of an amplifier in an electronic circuit? (ii) From ohm's law, show that the voltage gain is defined as

$$A_v = \frac{r_E}{r_C}$$

- (b) (i) Define gain in the context of an amplifier.
(ii) Based on the input and output signals, list the different types of amplifier.
(iii) State three (3) key parameters to consider when selecting an amplifier for specific application.
(iv) Explain the term "bandwidth" as it relates to amplifiers.
(v) What is the difference between voltage amplifier and current amplifier?

(17.5 mark)

QUESTION 6

1. Briefly describe the three ways to configure a transistor for amplification.

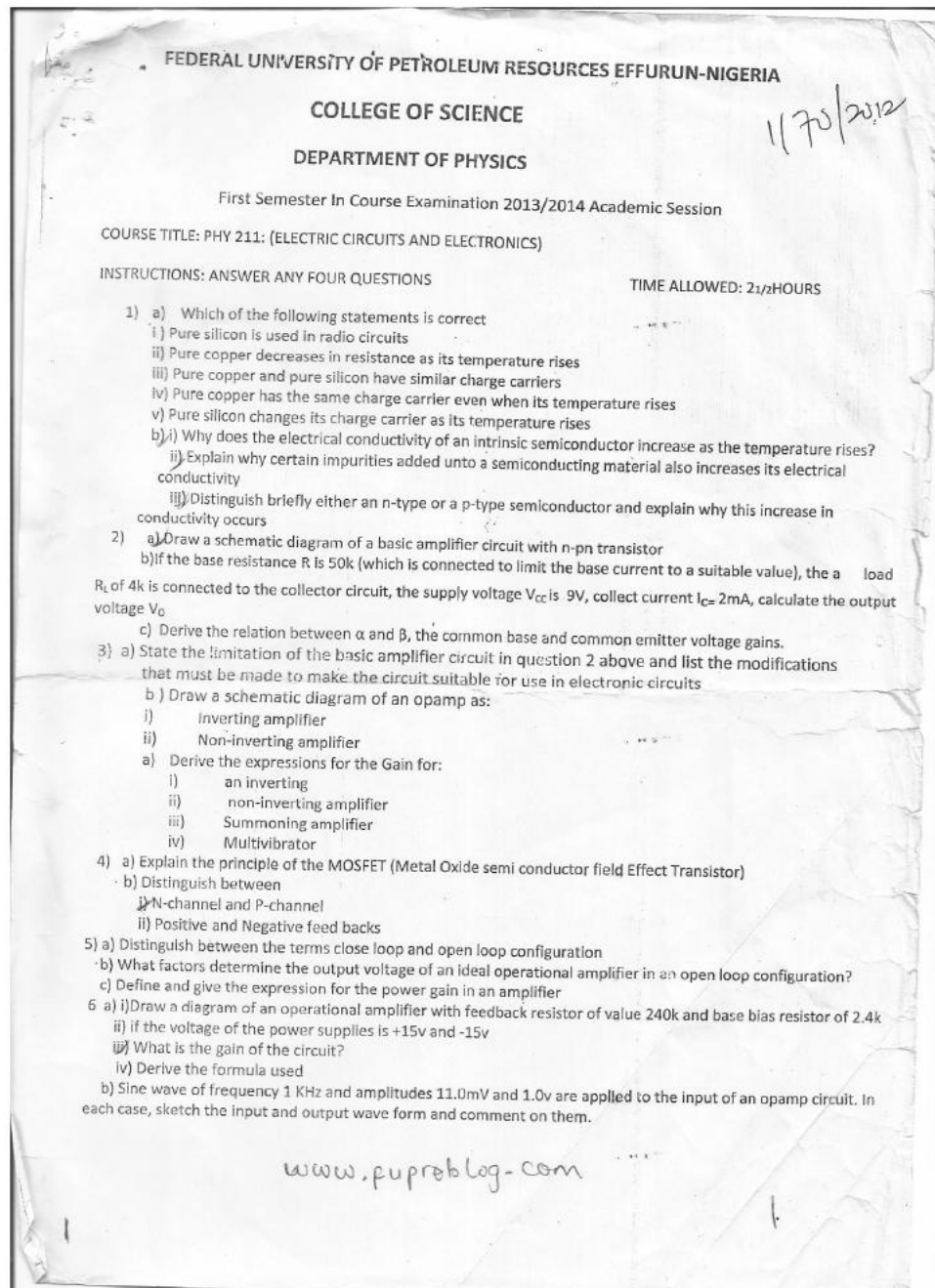
2. For a transistor amplifier, when the signal changes by 0.02V, the base current changes by 10 μA and collector current by 1mA. If the load $R_L = 10k\Omega$ and $R_C = 10k\Omega$, calculate the (i) current gain (ii) input impedance (ii) a.c. load (iii) voltage gain (iv) power gain.

(15 mark)

Paper 3 of 7 — PHY 211 2013 2014 EXAM

Level 200 • First Semester • PastQuestion

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Paper 4 of 7 — PHY 211 2016 2017 EXAM

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PYHSICS

COURSE TITLE: ELECTRIC CIRCUITS AND ELECTRONICS:

COURSE CODE: PHY 211

ACADEMIC SESSION: 2016/2017 TIME ALLOWED: 2 hours 30 minutes

INSTRUCTIONS: ANSWER ANY FOUR QUESTIONS

* Question 1

- Explain cut off Region and saturation point of a transistor.
- State and explain the two mechanisms responsible for reverse junction breakdown voltage.
- A transistor connected in a common base configuration has emitter current, $I_E = 5\text{mA}$, $I_B = 3.5\mu\text{A}$, (i) Determine the short circuit gain. (ii) Obtain an expression to show the relationship between the common emitter gain and the common base circuit gain.

Question 2

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Ze) A common emitter configuration has an output voltage $V_{CE} = 20\text{KV}$, load resistance $R_L = 20\text{K}\Omega$; $(h_{fe}) = 25$; $R_{in} = 10\text{K}\Omega$. (i) Calculate the input voltage V_{in} . (ii)

Output current I_C . (iii) Voltage gain A_v .

Question 3

2) Explain the differences between the direct band gap semiconductor and the indirect band gap semiconductor.

b) State four features of an N-type semiconductor, and draw the lattice structure of silicon doped with Arsenic.

A Gallium Phosphide LED with a band gap of 3.62eV is incident upon by a photon with energy equal to the band gap energy. Find the momentum of the photon. What colour of light will be emitted by this material?

Question 4

3) Write a short note on the non-linearity of the diode, using the voltage/ current characteristic graph.

ae State the purpose of the three terminals and the channel of the Field Effect Transistor (FET).

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State five applications of a diode.

©) A transistor with a common emitter configuration has a load resistance of $3.5k\Omega$, provides a voltage gain of 105, with an input resistance of $1.5k\Omega$.

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Paper 5 of 7 — PHY 211 (ELECTRIC CIRCUIT AND ELECTRONICS)Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

ST

1 SEMESTER EXAMINATION (2014/2015 SESSION)

PHY 211 (ELECTRIC CIRCUIT AND ELECTRONICS) (3 UNITS)

Instruction: Answer question 1 and any other three (3) questions Time: 2½hrs

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(b) How does avalanche voltage compare with the forward break over voltage
(c) Suppose two signals, one having input frequency of 3.500MHz and the other having an input frequency of 2.5MHz are applied to the input of the mixer. What are the output signal frequencies?
(d) Suppose a single semi-conductor for PV cell can produce 100 μ A of current in direct sunlight. How many of these would be necessary, if connected in parallel to produce 500 μ A
(e) What is the static forward current transfer ration in a device where a base current of 3mA produces a collector current of 25mA
(f) The acronym MOSFET stands for
(g) What is the current gain in a transistor where a change in the base current of 0.1 μ A results in a change in the collector current of 1 μ A
(h) The term common drain implies that
(i) Give three ways which an NPN transistor can be used
(j) If the rms AC input voltage to a circuit is 10Mv and the rms AC output voltage is 70Mv. What is the voltage gain? Assume the input and output impedances are identical.
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(b) With circuit diagrams and with anode and cathode symbology diagram. Explain
(i) Forward bias current flow
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3. (a) Explain Signal Amplifier
(b) Draw a circuit diagram to explain Envelope Detection as an application of signal
4. (a) Explain the principle of JFET (Junction Field Effect Transistor)
(b) With pictorial diagrams and schematic symbols. Explain

(i) NPN Transistors

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5. (a) Explain the basic FET circuits.

(b) With circuit diagram, explain the common source circuit configuration.

6. (a) Explain Signal Oscillators

(b) With the simple diagram of the basic concept of signal oscillator. explain briefly (i) the feedback

Paper 6 of 7 — PHY 211 PQ 2024

Level 200 • First Semester • PastQuestion

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Cost/sem/2021
Amade muna Peter

Federal University of Petroleum Resources Effurun, Nigeria.
 College of Science, Department of Physics.
 First Semester Examination

Academic Session: 2023/2024
 Course Title: Electric Circuit & Electronics
 Course Code: PHY211 (3 units)
 Time Allowed: 2½ hrs.

Date: /05/2024

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QUESTION 3

(a) Discuss the operation of the Full Wave Bridge Rectifier
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 (c) How does Zener diode act as a voltage stabilizer?
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Paper 7 of 7 — PHY 211 Past Question 2014/2015

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

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PHY 211 (ELECTRIC CIRCUIT AND ELECTRONICS) (3 UNITS)

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6. (a) Explain Signal Oscillators

(b) With the simple diagram of the basic concept of signal oscillator. explain briefly (i) the feedback